



RD350

If you are really interested, India's largest RD350 forum is at tinyurl.com/rxeqmv



Street control

A Timing Retard Eliminator is used on street legal sport bikes to limit throttle and prevent accidents



Traction Control

A throttle reducing safety feature in street bikes

Geek Life



grab a microscope and put the key into the ignition of a 2008 Yamaha YZF-R6 and see what happens.

The tachometer needle sweeps through its range while that little black box silently boots up: ambient temperatures are recorded, sensors feed signals to the ECU, kickstand up - check, gearbox in neutral - check, clutch lever depressed - check, and we now have current in the little red button that fires the beast to life. But wait, that's not all. The crank position sensor tells the ECU exactly where the piston is in relation to TDC (top dead centre) while the YCC-T or Yamaha Chip Control Throttle checks the position of the throttle, making sure you don't flood the engine if you have it pinned to the stop. And the YCC-I or the Yamaha Chip Controlled Intake is busy making sure the variable length intake trumpets are fully extended. And all of that happened from the time you turned the key to the time you moved your thumb to fire her up. The symphony of the electronic orchestra is yet to begin...

Like Yamaha's YCC-I, every motorcycle manufacturer has various technological trademarks registered to their name, some gaining more

popularity than the others, while yet others just work silently in the background.

All enhancing your motorcycling in some little way or another. Think back to the early 2000's when TVS launched the *Fiero*. At the time, the motorcycle was by far the most advanced Indian motorcycle. But did any one talk about the multiple map ignition system that extracted every bit of juice from that engine? Or for that matter - Bangalore-based two-stroke tuning house RDDreams offering in-house designed, multiple maps ignition kits for the legendary RD350? Not a word on the street. But circa 2007 and the whole world is talking about Suzuki's S-DMS or Suzuki Drive Mode Selector - the three-way switch on the handle bar that puts the fearsome Suzuki Hayabusa on a leash.

Let's go back to that motorcycle idling in the background and we'll make an attempt to delve deeper into its workings.

Perfect curves Ignition Maps

The ignition system, more commonly known as the CDI, is a finely-tuned piece of timing genius. It's got one job and one job alone, and that's to make sure a spark is delivered to the combustion chamber when the need arises; the cylinders are so needy, they grab a few hundred sparks each second at high RPM. And that's what the much talked about ignition maps essentially are, a graph of 'time

of spark' against 'engine speed'.

So, for example, at slower engine speeds, the spark is fired shortly before (in terms of position) the piston reaches the top, but as engine speed increases, the spark fires earlier, to give the mixture time to fully ignite. Sounds confusing doesn't it? Now, if mastering this wasn't difficult enough, research engineers discovered that at different engine loads (climbing a hill, selected gear, two passengers, etc.), a further change of the timing curves would ensure that the engine is using its full potential, making the maps three dimensional. So, while TVS used these maps to enable efficient combustion and increased fuel economy, tuners and the Japanese biggies took this to a different level as they extracted every last bit of potential the motorcycle could offer without having any manual adjustments made to them. Then Suzuki reverse-engineered this tech with their S-DMS. It spoiled boy racer silly by giving him an option to flick a switch, restrict the power on tap, and make him feel like he was on top of things. Safe? Yes. Fun? Most definitely not. But hey, it's an evolving world and fun need not be a dangerous sport, right?

Slippery when wet Traction control

Traction control is another rider-aid which was criticised by racing greats and serious motorcyclists. But it is actually making motorcycling a whole lot safer and has been a long time coming. Ducati is the only manufacturer that has



FLIP FLOP

There are only two kinds of motorcyclist: those who have fallen and those who shall fall. Check out Dani Pedrosa earning his stripes. Lined up alongside Suzuki rider John Hopkins, Pedrosa forgets to switch on the most important electronic launch control system on his Honda RC211V unleashing 250bhp instantaneously at the rear wheel. Oops!
<http://tinyurl.com/r3fsql>



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